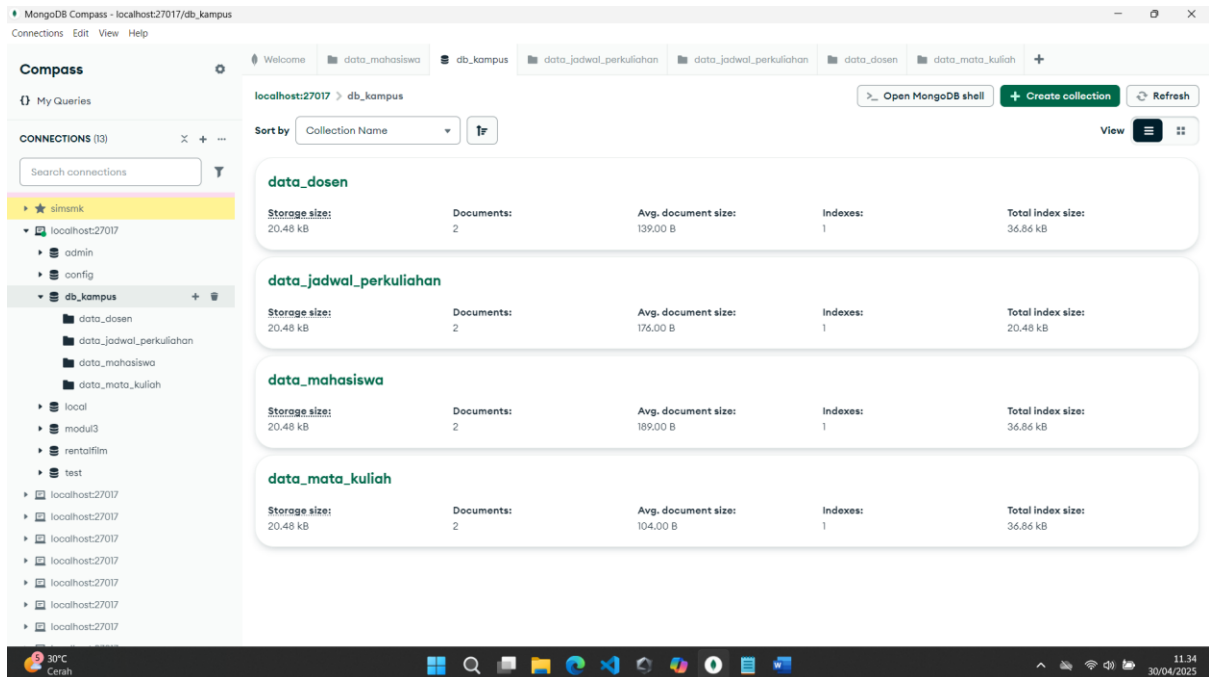


QUIZ 30 APRIL

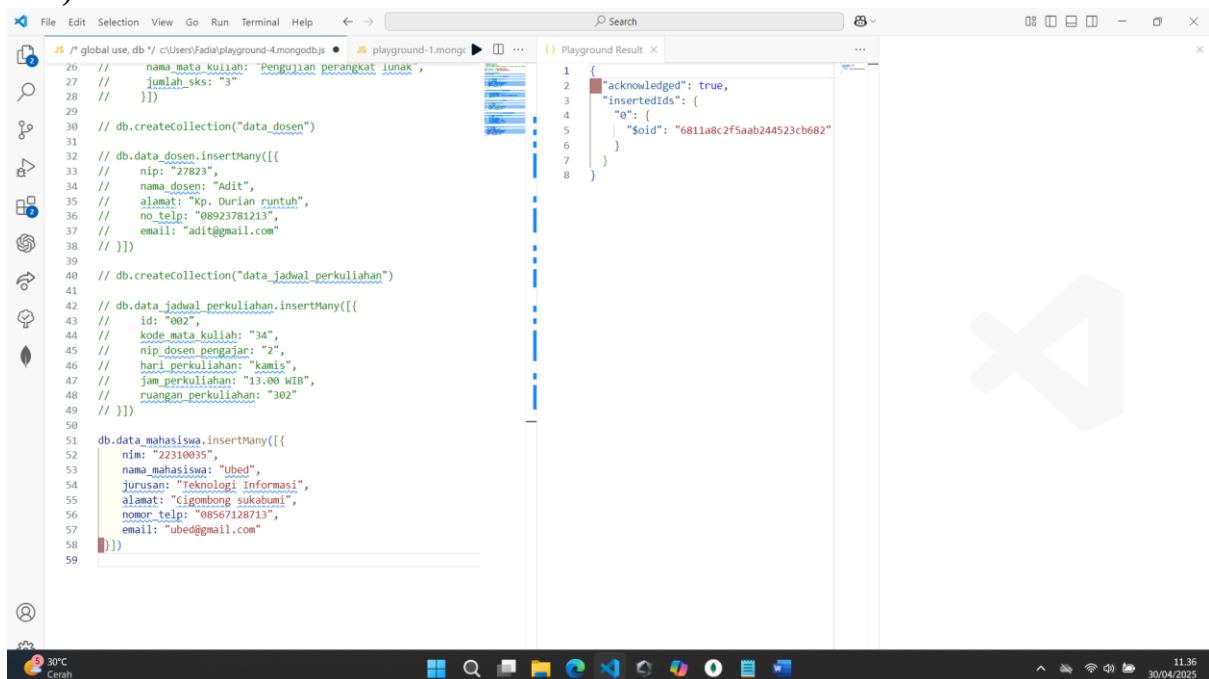
Anggota kelompok : Faiz Faqih Haidar (222310047)

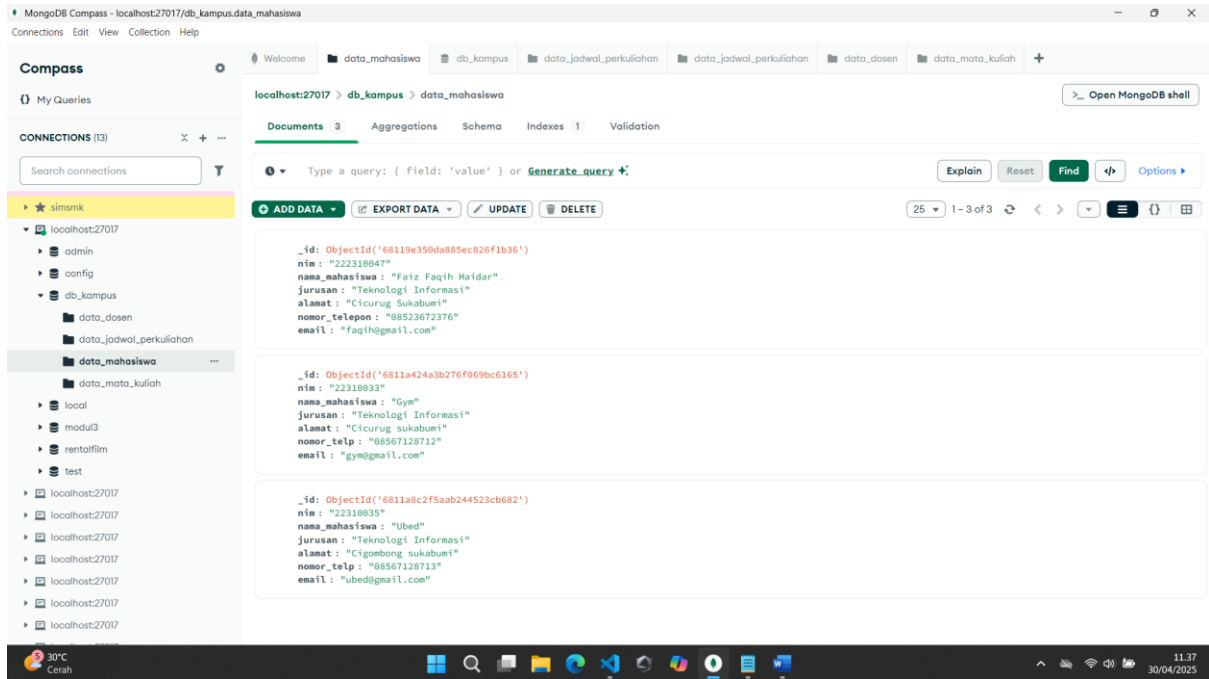
Muhamad Gymnastiar(222310007)

1

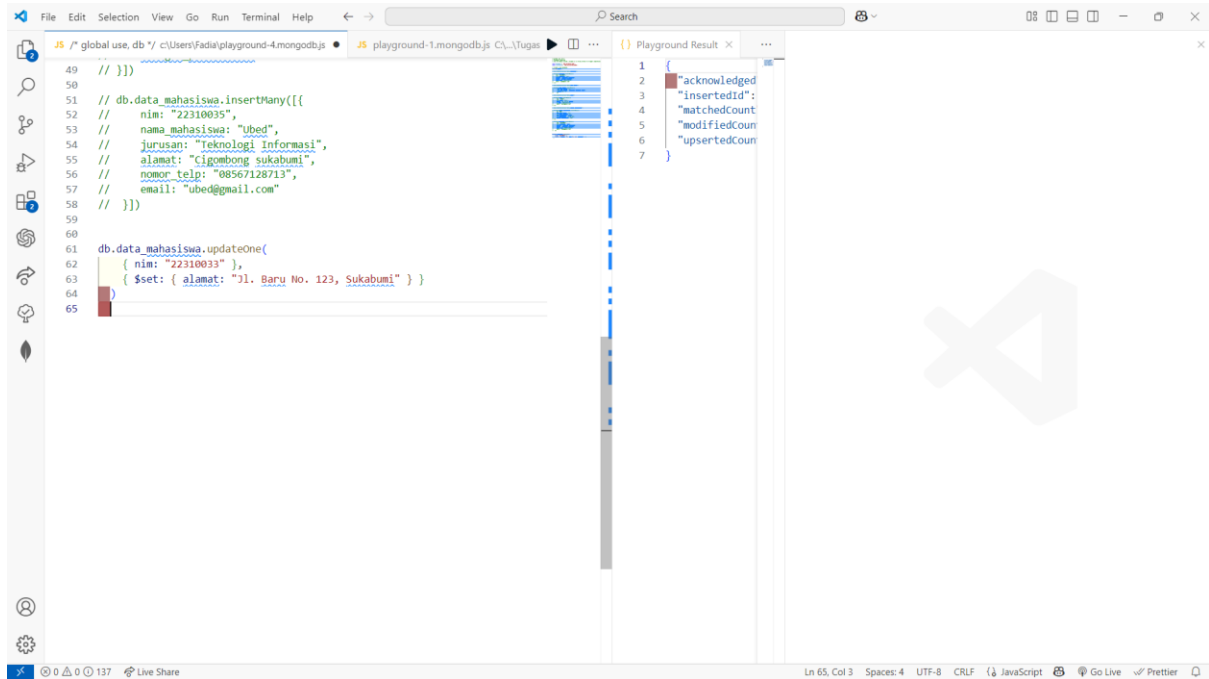


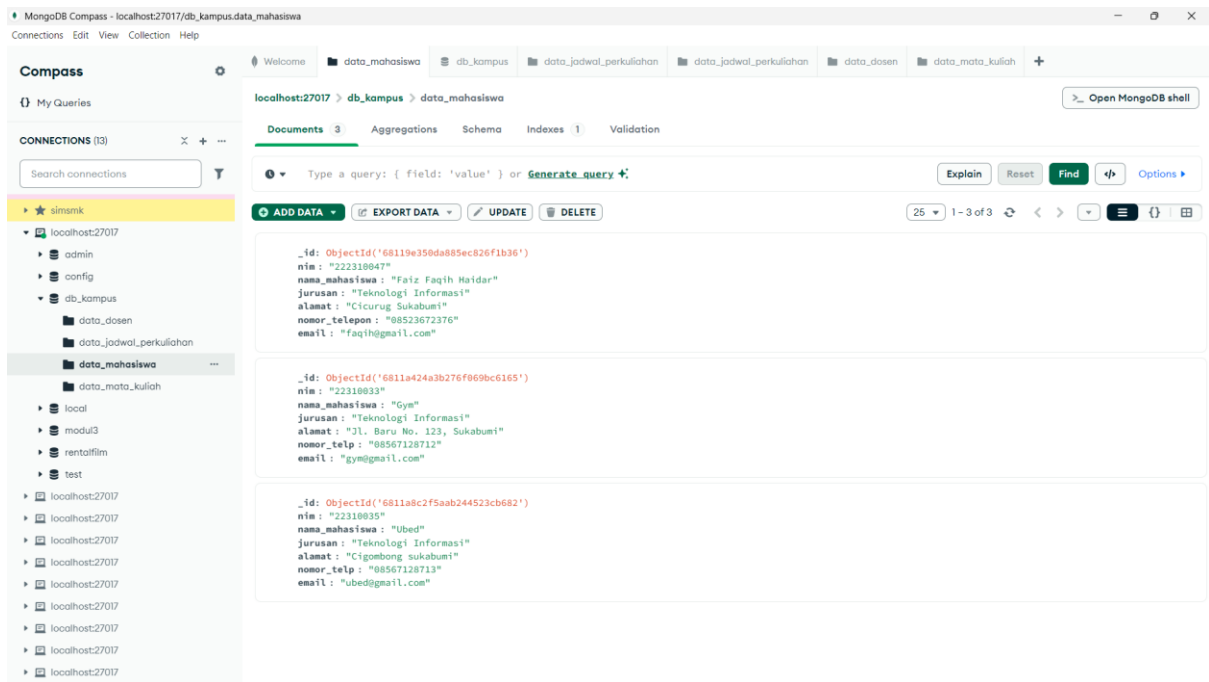
2. A)



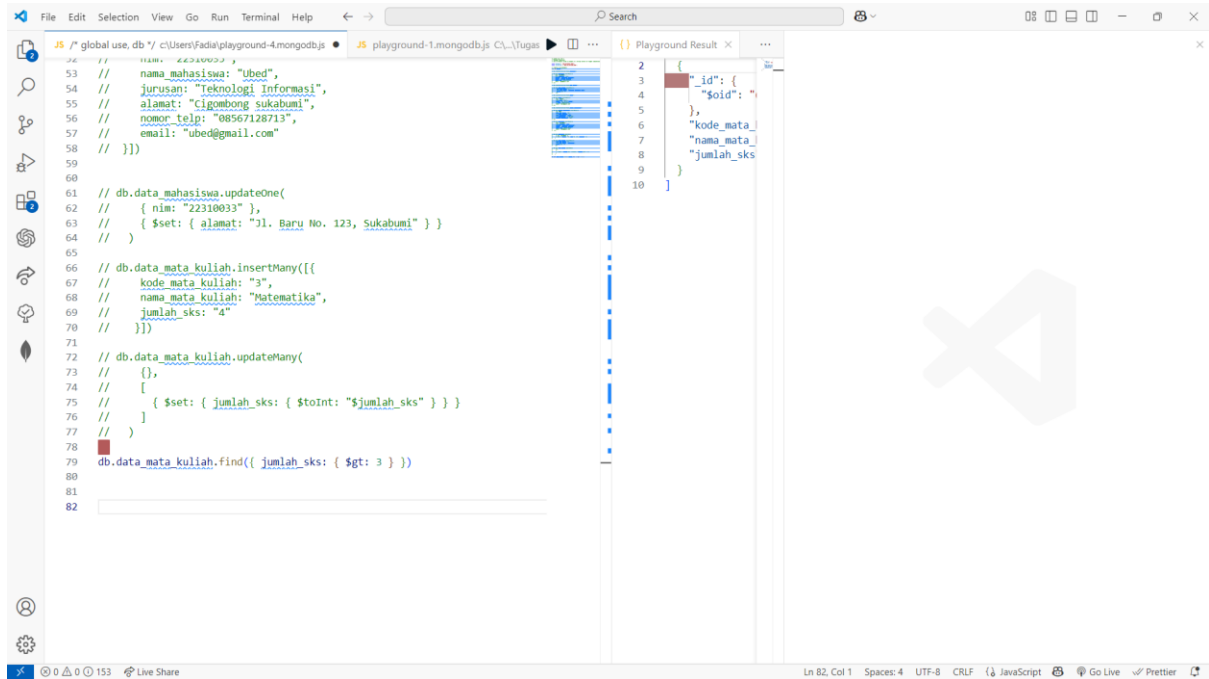


2. B)





2. C)



2. d)

2. E)

The image shows a development environment with VS Code and MongoDB Compass. The VS Code editor displays a JavaScript file named `playground-1.mongodb.js` with the following code:

```
60 // db.data_mahasiswa.updateOne(
61 //   { nim: "22310033" },
62 //   { $set: { alamat: "Jl. Baru No. 123, Sukabumi" } }
63 // )
64
65 // db.data_mata_kuliah.insertMany([
66 //   {
67 //     kode_mata_kuliah: "3",
68 //     nama_mata_kuliah: "Matematika",
69 //     jumlah_sks: "4"
70 //   }
71 // ])
72
73 // db.data_mata_kuliah.updateMany(
74 //   { },
75 //   { $set: { jumlah_sks: { $toInt: "$jumlah_sks" } } }
76 // )
77
78 // db.data_mata_kuliah.find({ jumlah_sks: { $gt: 3 } })
79
80 // db.data_dosen.insertOne(
81 //   {
82 //     nip: "27824",
83 //     nama_dosen: "Budi",
84 //     alamat: "Bandung",
85 //     no_telp: "082134567890",
86 //     email: "budi@gmail.com"
87 //   }
88 // )
89
90 db.data_dosen.updateOne(
91   { nip: "27823" },
92   { $set: { no_telp: "081234567890" } }
93 )
94
```

The MongoDB Compass interface shows the `data_dosen` collection with the following documents:

_id	nip	nama_dosen	alamat	no_telp	email
ObjectId('68119f690da885ec826f1b3e')	555512	Denis	Kp. durian runtuh	0856112892	denis@gmail.com
ObjectId('6811a5e3bde02d807054e53b')	27823	Adit	Kp. Durian runtuh	081234567890	adit@gmail.com
ObjectId('6811ad053b224a484d4fb46f')	27824	Budi	Bandung	082134567890	budi@gmail.com
ObjectId('6811ad1bad74d958e9952f')	27824	Budi	Bandung	082134567890	budi@gmail.com

2. F)

The image shows two screenshots of a development environment. The top screenshot is a Visual Studio Code editor window with a JavaScript file named `playground-1.mongod.js`. The code performs several MongoDB operations:

```
78 // db.data_mata_kuliah.find({ jumlah_sks: { $gt: 3 } })
79
80
81
82 // db.data_dosen.insertOne({
83 //   nip: "27824",
84 //   nama_dosen: "Budi",
85 //   alamat: "Bandung",
86 //   no_telp: "082134567890",
87 //   email: "budi@gmail.com"
88 // })
89
90 // db.data_dosen.updateOne(
91 //   { nip: "27823" },
92 //   { $set: { no_telp: "081234567890" } }
93 // )
94
95 db.data_jadwal_perkuliahan.deleteOne({ id: "002" })
96
```

The bottom screenshot is the MongoDB Compass interface. It shows the `data_jadwal_perkuliahan` collection in the `db_kampus` database. The `Documents` tab is active, displaying a single document:

```
{
  "_id": ObjectId("6811a96bda885ec826f1b47"),
  "id_jadwal": "001",
  "kode_mata_kuliah": "33",
  "nip_dosen_pengajar": "1",
  "hari_perkuliahan": "Rabu",
  "jam_perkuliahan": "10.00 WIB",
  "ruangan_perkuliahan": "301"
}
```

The interface also shows a sidebar with connections and a top bar with various toolbars for queries, documents, and schema management.

The screenshot shows a VS Code editor with a MongoDB playground script. The script is written in JavaScript and uses the MongoDB Node.js driver. It connects to a MongoDB instance, inserts a new document into the 'dosen' collection, updates an existing document, and creates an index on the 'kode_mata_kuliah' field. The results pane on the right shows the output of the script, which includes the inserted document and the updated document.

```

1 // global use, db 'c:\Users\Fadia\playground-4-mongodb.js'
2 // playground-1.mongodb.js C:\Users\Fugia 6
3
4 // db.data_mata_kuliah.find({ jumlah_sks: { $gt: 3 } })
5
6 // db.data_dosen.insertOne({
7 //   nip: "27824",
8 //   nama_dosen: "Budi",
9 //   alamat: "Bandung",
10 //   no_telp: "082134567890",
11 //   email: "budi@gmail.com"
12 // })
13
14 // db.data_dosen.updateOne(
15 //   { nip: "27823" },
16 //   { $set: { no_telp: "082134567890" } }
17 // )
18
19 // db.data_jadwal_perkuliahan.deleteOne({ id: "002" })
20
21 db.data_jadwal_perkuliahan.createIndex({ kode_mata_kuliah: 1 })
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96
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98
99

```

The results pane shows the output of the script, which includes the inserted document and the updated document.

```

1 kode_mata_kuliah
2
3
4
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16
17
18
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88
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92
93
94
95
96
97
98
99

```

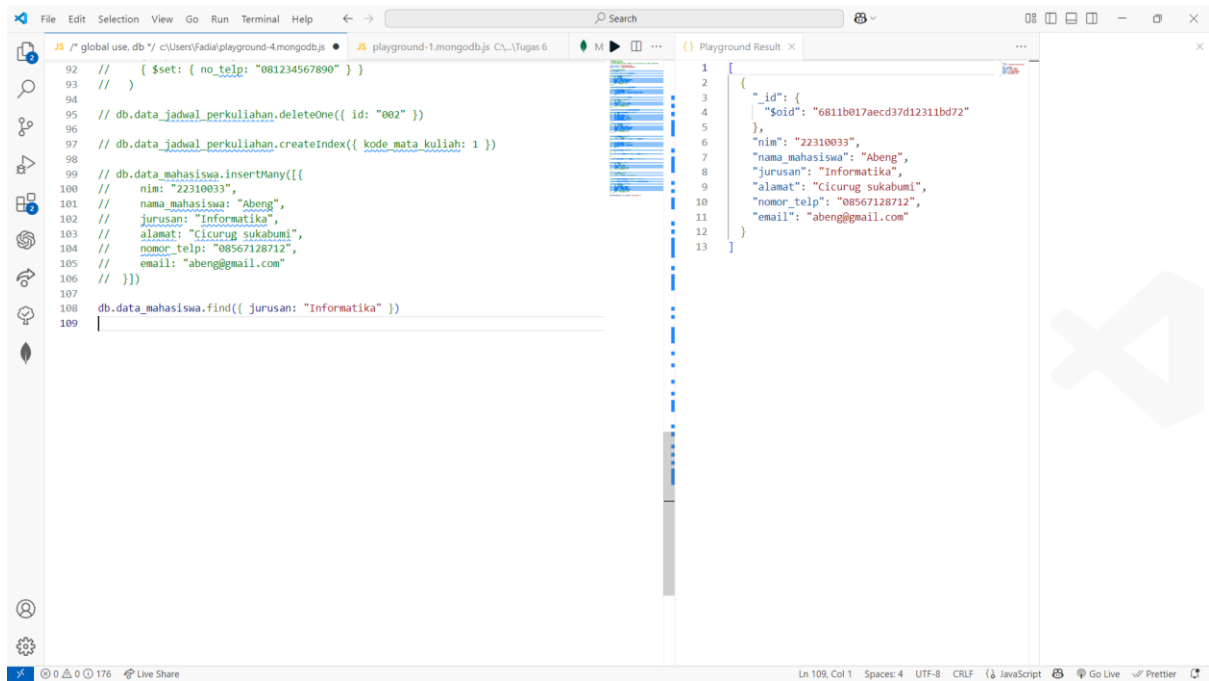
The screenshot displays the MongoDB Compass application. At the top, the title bar reads "MongoDB Compass - localhost:27017/db_kampus.data_jadwal_perkuluhan". Below the title bar, a menu bar includes "Connections", "Edit", "View", "Collection", and "Help".

The main interface is divided into several sections:

- Left Sidebar:** Contains a "Compass" header and a "My Queries" section. Below these are "CONNECTIONS (13)" and a search bar. A list of connections is shown, with "localhost:27017" selected. Under "localhost:27017", a tree view shows databases like "admin", "config", "db_kampus", "data_dosen", "data_jadwal_perkuluhan" (selected), "data_mahasiswa", "data_mata_kuliah", "local", "modul3", "rentaalfim", "test", and several other "localhost:27017" entries.
- Top Bar:** Shows the current database path: "localhost:27017 > db_kampus > data_jadwal_perkuluhan". There is an "Open MongoDB shell" button on the right.
- Documents Tab:** The "Documents" tab is active, showing a list of documents. The first document is displayed in a preview window:


```
{
  "_id": ObjectId("6811a06b0da885ec826f1b47"),
  "id_jadwal": "801",
  "kode_mata_kuliah": "33",
  "nip_dosen_pengajar": "1",
  "hari_perkuluhan": "rabu",
  "jam_perkuluhan": "16.00 WIB",
  "ruangan_perkuluhan": "301"
}
```
- Bottom Bar:** Includes buttons for "ADD DATA", "EXPORT DATA", "UPDATE", and "DELETE". It also shows a status bar with "25" documents, "1-1 of 1", and navigation icons.

4. a)



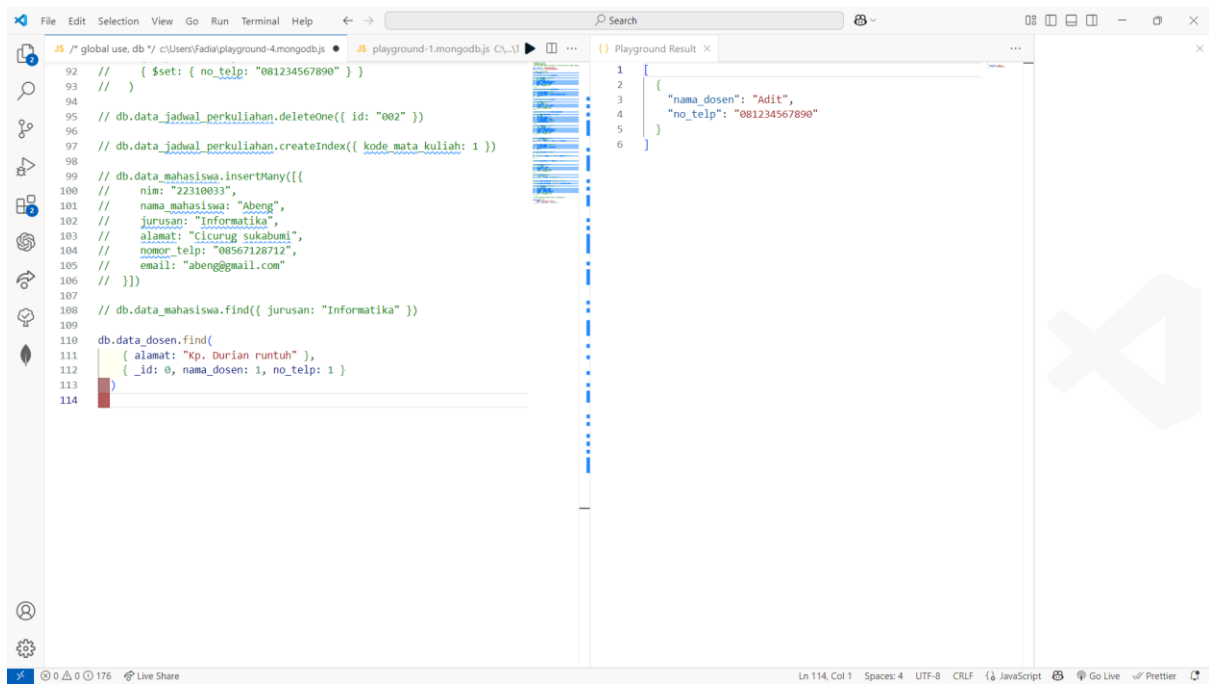
The screenshot shows a VS Code editor with a MongoDB Playground script. The script performs the following operations:

- Line 92: `{ $set: { no_telp: "081234567890" } }`
- Line 93: `{ }`
- Line 95: `db.data_jadwal_perkuliahan.deleteOne({ id: "002" })`
- Line 96: `db.data_jadwal_perkuliahan.createIndex({ kode_mata_kuliah: 1 })`
- Line 98: `db.data_mahasiswa.insertMany([`
- Line 99: `{`
- Line 100: `nim: "22310033",`
- Line 101: `nama_mahasiswa: "Abeng",`
- Line 102: `jurusan: "Informatika",`
- Line 103: `alamat: "Cicurug sukabumi",`
- Line 104: `nomor_telp: "08567128712",`
- Line 105: `email: "abeng@gmail.com"`
- Line 106: `})`
- Line 107: `])`
- Line 108: `db.data_mahasiswa.find({ jurusan: "Informatika" })`
- Line 109: `|`

The Playground Results pane shows the output of the `find` operation, displaying a single document:

```
{
  "_id": {
    "$oid": "6811b017aec37d12311bd72"
  },
  "nim": "22310033",
  "nama_mahasiswa": "Abeng",
  "jurusan": "Informatika",
  "alamat": "Cicurug sukabumi",
  "nomor_telp": "08567128712",
  "email": "abeng@gmail.com"
}
```

4. b)



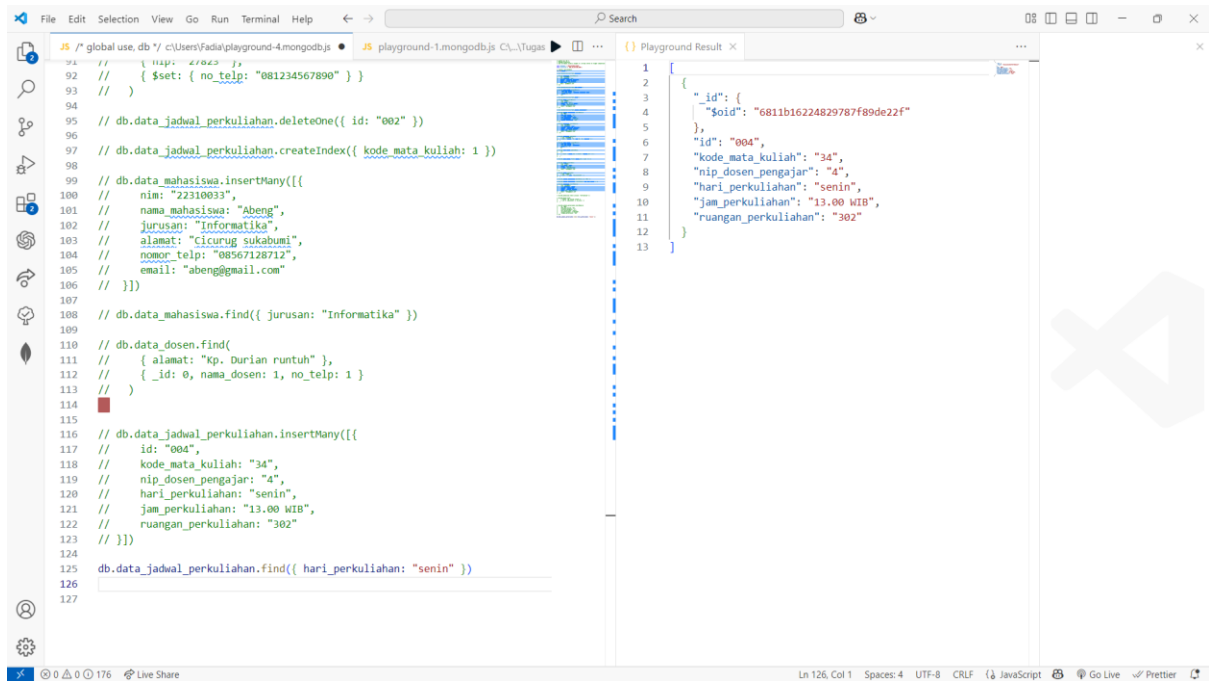
The screenshot shows a VS Code editor with a MongoDB Playground script. The script performs the following operations:

- Line 92: `{ $set: { no_telp: "081234567890" } }`
- Line 93: `{ }`
- Line 95: `db.data_jadwal_perkuliahan.deleteOne({ id: "002" })`
- Line 96: `db.data_jadwal_perkuliahan.createIndex({ kode_mata_kuliah: 1 })`
- Line 98: `db.data_mahasiswa.insertMany([`
- Line 99: `{`
- Line 100: `nim: "22310033",`
- Line 101: `nama_mahasiswa: "Abeng",`
- Line 102: `jurusan: "Informatika",`
- Line 103: `alamat: "Cicurug sukabumi",`
- Line 104: `nomor_telp: "08567128712",`
- Line 105: `email: "abeng@gmail.com"`
- Line 106: `})`
- Line 107: `])`
- Line 108: `db.data_mahasiswa.find({ jurusan: "Informatika" })`
- Line 109: `|`
- Line 110: `db.data_dosen.find(`
- Line 111: `{ alamat: "Kp. Durian runtu",`
- Line 112: `_id: 0, nama_dosen: 1, no_telp: 1 }`
- Line 113: `)`
- Line 114: `|`

The Playground Results pane shows the output of the `find` operation, displaying a single document:

```
{
  "nama_dosen": "Adit",
  "no_telp": "081234567890"
}
```


4. C)



The screenshot shows a VS Code editor with a MongoDB playground script on the left and its result on the right. The script is a JavaScript file named `playground-1.mongodb.js` located at `C:\Users\Fadiah\playground-4\mongodb.js`. The script performs several database operations: it sets a `no_telp` field to `"081234567890"` for a document with `_id: "002"`; it deletes a document with `_id: "002"`; it creates an index on the `kode_mata_kuliah` field of the `db.data_jadwal_perkuliahan` collection; it inserts a new student document into the `db.data_mahasiswa` collection; it finds all documents in the `db.data_dosen` collection where the `alamat` is `"Kp. Durian runtuh"`; it finds all documents in the `db.data_jadwal_perkuliahan` collection where the `hari_perkuliahan` is `"senin"`; and it inserts a new lecture document into the `db.data_jadwal_perkuliahan` collection.

```
1 /* global use, db */
2 c:\Users\Fadiah\playground-4\mongodb.js
3
4 // { no_telp: "081234567890" }
5 //
6 // db.data_jadwal_perkuliahan.deleteOne({ id: "002" })
7 // db.data_jadwal_perkuliahan.createIndex({ kode_mata_kuliah: 1 })
8 // db.data_mahasiswa.insertMany([
9 //   {
10 //     nim: "22310033",
11 //     nama_mahasiswa: "Abeng",
12 //     jurusan: "Informatika",
13 //     alamat: "Cicurug sukabumi",
14 //     nomor_telp: "08567128712",
15 //     email: "abeng@gmail.com"
16 //   }
17 // ])
18 // db.data_mahasiswa.find({ jurusan: "Informatika" })
19 // db.data_dosen.find(
20 //   { alamat: "Kp. Durian runtuh" },
21 //   { _id: 0, nama_dosen: 1, no_telp: 1 }
22 // )
23 // db.data_jadwal_perkuliahan.insertMany([
24 //   {
25 //     id: "004",
26 //     kode_mata_kuliah: "34",
27 //     nip_dosen_pengajar: "4",
28 //     hari_perkuliahan: "senin",
29 //     jam_perkuliahan: "13.00 WIB",
30 //     ruangan_perkuliahan: "302"
31 //   }
32 // ])
33 db.data_jadwal_perkuliahan.find({ hari_perkuliahan: "senin" })
```

The result of the script is a JSON document with the following fields:

```
1 {
2   "_id": {
3     "$oid": "6811b16224829787f89de22f"
4   },
5   "id": "004",
6   "kode_mata_kuliah": "34",
7   "nip_dosen_pengajar": "4",
8   "hari_perkuliahan": "senin",
9   "jam_perkuliahan": "13.00 WIB",
10  "ruangan_perkuliahan": "302"
11 }
12
13
```

The status bar at the bottom indicates the file is a JavaScript file, the encoding is UTF-8, the line ending is CRLF, and the editor is using the Prettier formatter.